



**QUEEN'S
UNIVERSITY
BELFAST**

MSc Artificial Intelligence

Knowledge Engineering

20 November 2024



Introduction to Probabilistic Reasoning

- Logic and Knowledge Graphs are about statements of fact
- Do we always have complete knowledge about the world?
- Are our “facts” always precise?
- Logic can handle incomplete knowledge
- It cannot handle uncertain knowledge

Recap

- You have previously studied some elementary forms of probabilistic reasoning
- We are now going to build up the **Bayesian Network**, a probabilistic representation that is an attempt to model the *causal relationships* between a set of random variables
- We will first need to remind ourselves of some basic concepts from probability theory

The Joint Distribution

Distribution of multiple variables.

$$P(\underline{X}) = P(X_1, X_2, \dots, X_N)$$

↑
Vector

Prob that $(\overset{\text{var}}{X_1} = \overset{\text{value}}{x_1}) \wedge (X_2 = x_2) \wedge \dots \wedge (X_N = x_N)$

$$= P(X_1 = x_1, X_2 = x_2, \dots) \equiv P(\underline{X} = \underline{x})$$

$$= P(x_1, x_2, \dots, x_N) \equiv P(\underline{x})$$

X_2

	H	T
X_1		
H	0.5	0.2
T	0.2	0.1

$$\sum_{i,j} P(x_i, x_j) = 1$$

$$P(x_i, x_j) \geq 0 \quad \forall i, j$$

The Sum Rule

Distributions of subsets of variables from the J.D.

For example $P(X_j, X_k) = \sum_{i \neq j, k} P(X_1, X_2, \dots, X_N)$ (Integral for C.D.)

		X_2	
		H	T
X_1	H	0.5	0.2
	T	0.2	0.1

$$P(X_1) = \begin{array}{c|c} H & 0.7 \\ \hline T & 0.3 \end{array}$$

Marginalisation

Conditional Probability

Distribution of a subset of variables given fixed values of another subset.

$$P(X_i | X_j) = \frac{P(X_i, X_j)}{P(X_j)}$$

↑
"given"

		X_2	
		H	T
X_1	H	0.5	0.2
	T	0.2	0.1

$$P(X_1 | X_2 = H)$$

$$P(X_1 | X_2 = H) = \frac{P(X_1, X_2 = H)}{P(X_2 = H)}$$

$$P(X_1, X_2 = H) = \frac{H | 0.5}{T | 0.2}$$

$$P(X_2 = H) = 0.5 + 0.2 = 0.7$$

$$P(X_1 | X_2 = H) = \frac{H | 0.5 / 0.7}{T | 0.2 / 0.7} = \frac{0.714}{0.286} = 1$$

The Product Rule

$$\underbrace{P(X_i, X_j)}_{\text{Two vars}} = \underbrace{P(X_i | X_j) P(X_j)}_{\text{Univariate}}$$

Rearrangement of expression for conditional distribution

Factorisation of the joint distribution.

Independence

If distribution of X_i is independent of other variables

Then it is independent: $P(X_1, X_2, \dots, X_N) = P(X_i) P(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_N)$

If all variables are independent

$$P(X_1, \dots, X_N) = \prod_{i=1}^N P(X_i)$$

J.D Factorises.

Bayes Rule

$$P(X_i, X_j) = P(X_i | X_j) P(X_j)$$

$$P(X_j, X_i) = P(X_j | X_i) P(X_i)$$

$$P(X_i, X_j) = P(X_j, X_i) \quad [\text{J. D is symmetric}]$$

$$\therefore \frac{P(X_i | X_j) \cancel{P(X_j)}}{P(X_i)} = \frac{P(X_j | X_i) \cancel{P(X_i)}}{P(X_j)}$$

Bayes' Rules

Exercise: Probably just about OK



**QUEEN'S
UNIVERSITY
BELFAST**

MSc Artificial Intelligence

Knowledge Engineering

25 November 2024



Introduction to Probabilistic Reasoning

- Logic and Knowledge Graphs are about statements of fact
- Do we always have complete knowledge about the world?
- Are our “facts” always precise?
- Logic can handle incomplete knowledge
- It cannot handle uncertain knowledge

Recap

- You have previously studied some elementary forms of probabilistic reasoning
- We are now going to build up the **Bayesian Network**, a probabilistic representation that is an attempt to model the *causal relationships* between a set of random variables
- We will first need to remind ourselves of some basic concepts from probability theory

The Joint Distribution

The Sum Rule

Conditional Probability

The Product Rule

Independence

Bayes Rule

Exercise 13: Basic Probability

Conditional Independence

- Independence is a powerful and useful assumption in many problems
- But it does not apply in all situations
- Joint distribution is hard to work with in large problems
- Can we give the joint distribution some structure akin to independence but more general?

Conditional Independence

Conditional Independence

Conditional Independence

Exercises 14: Conditional Independence

A Computational Example

- Let us now study conditional independence computationally via an example
- Pedagogical exercise - not how you would do this in real-world problems

The Joint Distribution

- Survey of school-leavers' employment ten years after they finished high school.
- The survey look at their careers:
 - which sector they worked in and what type of work they did)
 - and the subjects in which they graduated from high school.
- We will consider a small part of this survey data and will look at the subset of people who
 - Studied Mathematics and/or Geography (or neither)
 - Pursued careers in Programming and/or Finance (or neither)

In other words, the full joint distribution has been marginalised over all variables except

- M : studied Mathematics.
- G : studied Geography.
- F : works in finance.
- P : works as a programmer.

The Joint Distribution

		M		$\neg M$	
		P	$\neg P$	P	$\neg P$
G	F	0.099	0.081	0.009	0.011
	$\neg F$	0.066	0.054	0.036	0.044
$\neg G$	F	0.1485	0.1215	0.0135	0.0165
	$\neg F$	0.099	0.081	0.054	0.066

Can you discern any obvious structure?

Any Patterns?

- Not obvious what conclusions we can draw from this.
- Is there any substructure in this from which we might infer something about the relationship between the variables?
- We can calculate things from the joint distribution to see whether there are any patterns

`Open Conditional Independence nocode.ipynb`

Conditional Independence

What did we find?

- Careers in finance and programming don't depend on having studied German
- They do depend on having studied Maths
- But they don't depend on each other.
- How does this help us?
- It allows us to **factorise** the joint distribution

The Product Rule Revisited

Exercise 15: Conditional Independence 2



**QUEEN'S
UNIVERSITY
BELFAST**

MSc Artificial Intelligence

Knowledge Engineering

25 November 2024



Bayesian Networks

- Last lecture, established the idea of *conditional independence*

$$P(X, Y \mid Z) = P(X \mid Z) P(Y \mid Z)$$

- We showed that the product rule could be used to factorise the joint distribution in terms of conditional probabilities, for example

$$P(G, M, F, P) = P(F \mid M) P(P \mid M) P(M) P(G)$$

- Why is this useful and desirable?

Benefits of Factorisation

1. Reduced number of parameters

$$P(G, M, F, P) = P(F | M) P(P | M) P(M) P(G)$$

15 parameters *6 parameters*

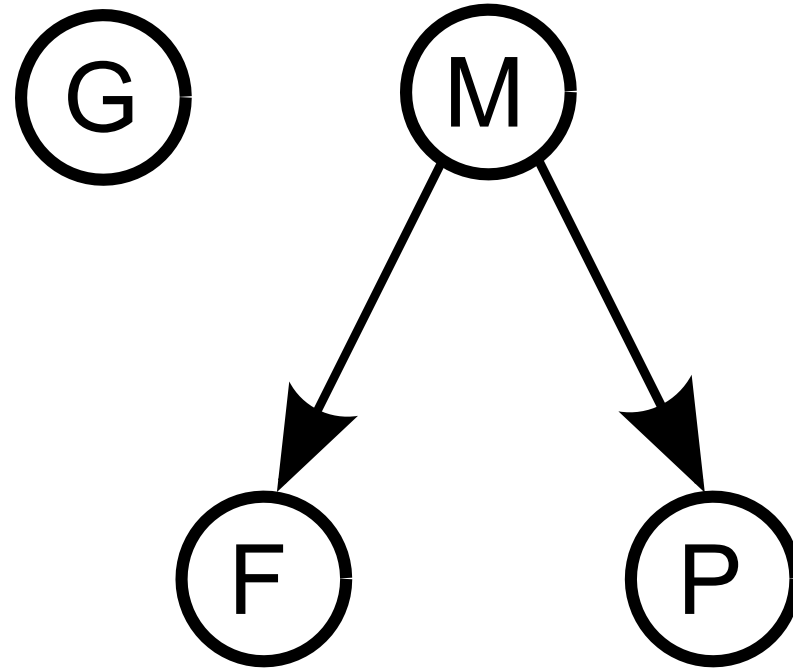
2. Univariate conditional PDFs much easier to measure/estimate
3. Explicitly models causal structure

Models with the structure are called **Bayesian Networks**

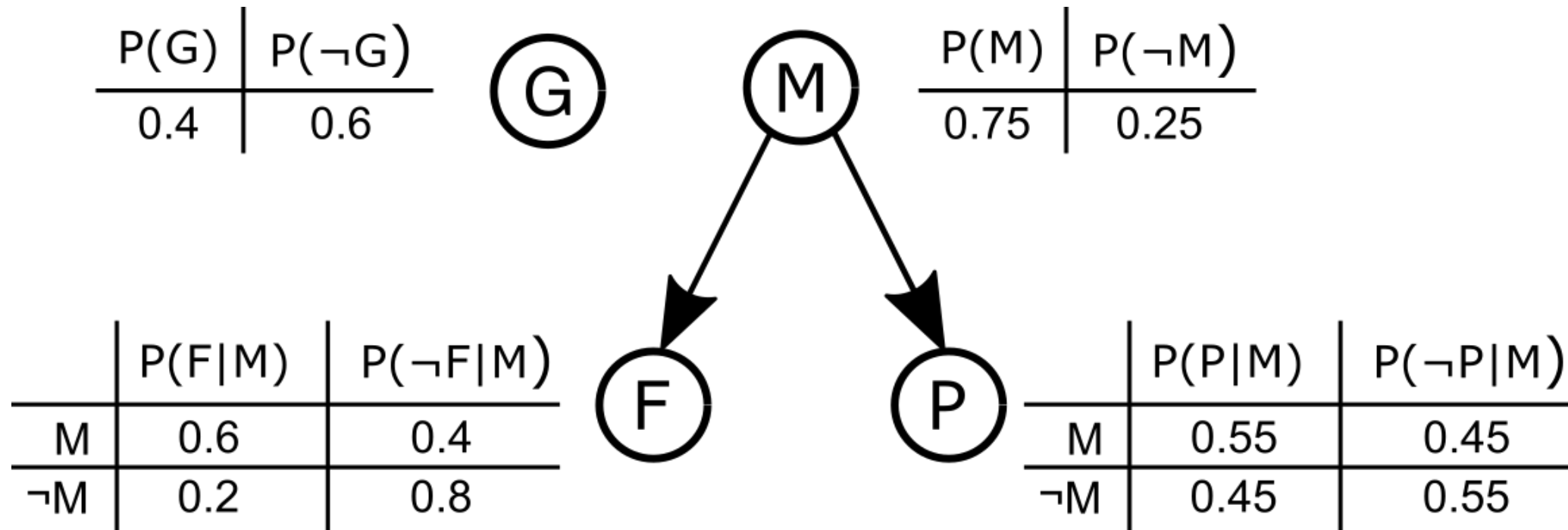
Formal Definition

- Graphical representation of factorised distribution
- Directed graph
 - Each node corresponds to a single variable
 - Edges between node represents conditional relationships
 - Each node has associated condition probability distribution
 - Strictly acyclic – a directed acyclic graph (DAG)

Example



Example



Constructing a DAG

1. Choose the variables needed to model the domain and order them. A good order has causes preceding effects and this will lead for a more compact representation.
2. For each node, choose a set of parents such that the chain rule is satisfied and that the parents can plausibly represent causes of the node's effect.
3. Add edges directed from parents to the node.
4. Write down a conditional probability table for the node.

Example

Consider the following four events

- Fire brigade called
- Cat stuck in tree
- House on fire
- Explosion in kitchen

What might a DAG for these look like?



Exercise 16: Bayesian Networks



**QUEEN'S
UNIVERSITY
BELFAST**

MSc Artificial Intelligence

Knowledge Engineering

28 November 2023



Inference in Bayesian Networks

- Once we have a Bayesian network, what can we do?
- Network represents the joint distribution
- So we can do anything that we can do with the joint distribution
- Can compute the joint distribution and derive quantities from that
- But that is wasteful and negates the advantages of the network
- Can we compute directly?

Inference in Bayesian Networks

- Open notebook

9. Exact Inference in Bayesian Networks `nocode.ipynb`

Exercise: Exactly



**QUEEN'S
UNIVERSITY
BELFAST**

MSc Artificial Intelligence

Knowledge Engineering

25 November 2024



Approximate Inference

Approximate Inference

Approximate Inference



**QUEEN'S
UNIVERSITY
BELFAST**

MSc Artificial Intelligence

Knowledge Engineering

25 November 2024



Recap

- Yesterday we showed how to perform exact computations on Bayesian Networks
- And how to use random sampling approaches to perform approximate inference.
- Both approaches have disadvantages
- Exact inference is laborious and challenging
- Discrete random sampling is slow, and also wasteful (Why?)
- Today – two methods for improving approximate inference

Importance Sampling

- Discrete sampling indiscriminately samples the whole sample space
- Can support multiple queries
- But for answer a specific question, very inefficient
- Imagine we want to know $P(M \mid P=\text{True}, G=\text{False})$.
- We can sample the whole space and use rejection sampling
- Many samples go unused...
- Can we just sample the subset that we want?
- Yes (importance sampling)
- Can be made even more efficient – *likelihood weighting*.

Likelihood weighting

- Imagine we have some evidence that means some variables should be fixed – *evidence variables*
- The evidence must by definition be consistent with the posterior distribution
- Sampling from the sample subspace that supports the evidence must take this into account
- Not all evidence is equally well supported – we should account for the probability that a sample is more or less supportive of the evidence
- This is the principle behind likelihood weighting

The Likelihood Weighting Algorithm

- Fix all the evidence variables to their observed values
- Sample non-evidence variables conditioned on the fixed evidence variables
- Compute the probability that the sample you have just drawn would give rise to the evidence that you have fixed.
- Include your sample in the trace, weighted by this probability.

Example

Imagine we want to know $P(M \mid P=\text{True}, G=\text{False})$.

- Fix $P = \text{True}$.
- Set weight $w = 1.0$.
- G is an evidence variable and is fixed at False. Set $w = w \times P(G=\text{False}) = 0.4$.
- M is not evidence. Sample from $P(M) = (0.25, 0.75)$, suppose this returns True.
- Sample from $P(F \mid M=\text{True}) = (0.4, 0.6)$, suppose this returns False.
- P is an evidence variable and is True. Set $w = w \times P(P=\text{True} \mid M=\text{True}) = 0.4 \times 0.55 = 0.22$.
- Record (False, True, False, True) with weight 0.22.

Why does this work?

- All of the samples generated have the observed values of the evidence variables and so all of the samples can be used.
- Improved accuracy (more samples) or faster convergence for a given level of accuracy.
- Can be shown rigorously that weighting of the likelihoods in this way is fully consistent with direct sampling from the desired posterior distribution.

Markov Chain Monte Carlo

- Discrete sampling and its variants all generate samples independently.
- A new sample in the trace has no knowledge of any previous samples that have been generated.
- A very common alternative treats the sampling as a *Markov Chain* in which the generation of the next sample depends on the current sample, but not on events prior to the current sample.
- Markov Chain Monte Carlo (MCMC) sampling methods generate each successive sample by randomly modifying the current sample.
- Gibbs Sampler is especially well suited for Bayesian Networks.

Gibbs Sampling

- The Gibbs sampler is a Markov Chain
- States are generated by making modifications to the current state
- Begin by fixing all evidence variables to their observed values. These will not be changed through the sampling process.
- Initialise all of the non-evidence variables to random values.
- Initial values must be supported by the distribution attached to the variable but there are no other requirements.
- A new state is created by sampling **one** of the non-evidence variables, chosen at random.
- The sampling of the chosen variable will be from its conditional probability distribution which is implicitly or explicitly conditioned on its *Markov Blanket*

Markov Blanket

- The set of variables that are either parents or children of the chosen variable, or are co-parents of the variable's children
- Any variable is conditionally independent of all variables outside of its Markov blanket

Gibbs Sampling

- The Gibbs sampler determines its outcome by counting how many times it visits each state and normalising.
- If it visits a state where M is True forty-five times, and M is False fifteen times, the resulting distribution is $P(M \mid F)$
- The Gibbs sampler settle into an equilibrium in which it visits states according to a distribution that is equivalent to the posterior
- Needs to reach that equilibrium – requires a *burn-in* period
- Tricky to implement – we use a library: `pymc3`
- Open 11. Approximate Inference `nocode.ipynb`